Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024

Course: B.Tech
Code: 100111

Subject: Programming for Problem Solving

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is a valid variable name in C?
(i) 2value (ii) @value
(iii) value_2 (iv) value#2

(b) Which of these is not a valid data type in C?
(i) int (ii) real
(iii) float (iv) char

(c) Which of the following is the correct syntax for if statement?
(i) if [condition] then (ii) if condition then
(iii) if (condition) (iv) if {condition}

(d) What is the output of printf("%d", 5 + 3 * 2);?
(i) 16 (ii) 10
(iii) 11 (iv) 13

(e) Which header file is required to use printf()?
(i) stdlib.h (ii) stdio.h
(iii) conio.h (iv) string.h

(f) Which keyword is used to define a constant in C?
(i) final (ii) #define
(iii) constant (iv) static

(g) What is the index of the first element in an array?
(i) 1 (ii) -1
(iii) 0 (iv) Depends on array

(h) Which one is the logical AND operator in C?
(i) & (ii) |
(iii) && (iv) ||

(i) Which format specifier is used for printing float values?
(i) %d (ii) %c
(iii) %f (iv) %x

(j) Which of these is not a loop in C?
(i) for (ii) do-while
(iii) while (iv) repeat-until

Q.2 (a) Explain the structure of a C program with an example. [7]
(b) Describe basic data types in C with memory size. [7]

Q.3 (a) What is an algorithm. Write an algorithm to perform the addition of two numbers in C and draw a flowchart to represent the addition of two numbers. [7]
(b) Write a program to check whether a number is even or odd using if-else. [7]

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Q.4	(a)	Explain Arithmetic and Logical operators in C with example programs.	[7]
	(b)	Explain Relational and Bitwise operators with example programs.	[7]
Q.5	(a)	Explain call by value and call by reference with examples.	[7]
	(b)	What is the difference between library functions and user-defined functions?	[7]
Q.6	(a)	What is an array in C programming? Write advantages of using arrays. Explain a one-dimensional array with a suitable example program.	[7]
	(b)	Describe the difference between character array and string.	[7]
Q.7	(a)	Write the difference Between Array and Structure in C programming.	[7]
	(b)	Write the difference Between Structure and Union in C programming.	[7]
Q.8	(a)	What is a pointer? Explain pointer declaration and initialization with sample program.	[7]
	(b)	What is file? Explain text file and binary file in details.	[7]

Q.9 Write short notes on ***any two*** of the following:- **[7 x 2 = 14]**

(a) Structure
(b) Union
(c) Recursion
(d) Fibonacci series program

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